Conjectural Grothendieck–Teichmüller Generators Act Nontrivially on Quadratic Poisson Structures

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Abstract

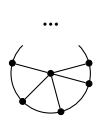
For every odd $N \geq 5$, we prove that, in Kontsevich's graph complex equipped with the edge-contraction differential, the wheel graph W_N is homologous to an explicit linear combination of $2^{(N-3)/2}$ graphs, each having only 3- and 4-valent vertices. Combined with the Khoroshkin–Merkulov construction of the action of the even graph complex on finite-dimensional quadratic Poisson structures [2], this implies that the odd wheel classes act non-trivially on such structures.

1 Introduction

By the results of Khoroshkin and Merkulov on deformation quantization of quadratic Poisson structures [2], the even graph complex acts on finite-dimensional quadratic Poisson structures. However, the faithfulness of this action is not established there. More precisely, a cohomology class acts non-trivially on quadratic Poisson structures if and only if it does not vanish under the projection

$$p\colon H(\mathsf{GC}_2, \delta) \to H\big(\mathsf{GC}_2/\langle \mathsf{GC}_2\rangle_{\geq 5}, \delta\big). \tag{1}$$

Here $\langle \mathsf{GC}_2\rangle_{\geq 5}$ denotes the differential closure of the subspace spanned by graphs having a vertex of valence at least 5. Thus, to prove nontriviality of a class, it suffices to show that it does not vanish under the projection (1). While it is conjectured that the projection (1) is a quasi-isomorphism, very few classes have been shown not to vanish under it. In this paper we consider the odd wheel classes

$\mathfrak{w}_N =$  $+ \cdots,$

and show that they remain nonzero under the projection (1), and thus act non-trivially on finite-dimensional quadratic Poisson structures.

We prove this by constructing explicit representatives of the corresponding homology classes in the (pre)dual graph complex, involving only graphs whose vertices have valence 3 or 4.

The odd wheel classes $[\mathfrak{w}_N]$ occupy a distinguished place in the graph complex. By Willwacher's identification of the degree-zero cohomology of Kontsevich's graph complex with the Grothendieck–Teichmüller Lie algebra [4], they correspond to the conjectural generators of $\mathfrak{grt}_1$ arising in the Deligne–Ihara conjecture. Brown showed that the free Lie algebra generated by the odd generators embeds into $\mathfrak{grt}_1$ [1].

2 Reformulation to the Dual Perspective

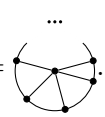
Instead of working with the vertex-splitting differential δ, we shall work with the edge-contraction differential d. From this (pre)dual perspective, the cohomological statement above becomes a homological statement about low-valence representatives.

Conjecture 2.1. *Let $(\widetilde{\mathsf{GC}_d})_{<5}$ be the maximal subcomplex of (GC_d, d) spanned by linear combinations of graphs all of whose vertices have valence 3 or 4. Then the inclusion*

$$i\colon ((\widetilde{\mathsf{GC}_d})_{<5}, d) \hookrightarrow (\mathsf{GC}_d, d)$$

is a quasi-isomorphism.

With the edge-contraction differential, the homology classes corresponding to the $\mathfrak{grt}$ generators may be represented by single wheel graphs W_N:

$W_N =$  .

However, for $N > 3$, the wheel graph W_N does not lie in the subcomplex $(\widetilde{\mathsf{GC}_d})_{<5}$. In this paper we prove the following theorem.

Theorem 2.2. *For d even and each odd $N \geq 5$, there exists an element $U_N \in \mathsf{GC}_d$ such that*

$$W_N - d(U_N) \in (\widetilde{\mathsf{GC}_d})_{<5}.$$

Theorem 2.2 is the homological statement dual to the nonvanishing of $p([\mathfrak{w}_N])$.

3 Conventions

We shall assume familiarity with Kontsevich's graph complex in this paper; for a formal definition and further discussion, see [3]. To fix conventions, we will consider GC_d to consist of connected graphs in which every vertex is at least 3-valent.

We will use vertex and edge labels starting from 0. We will exclusively consider the case that d is even. In this case, vertex labels and edge directions do not matter, and edge labels matter only through the sign orientation.

The differential d is the edge-contraction differential. If a graph has m edges, then the contraction of the edge labelled i carries the sign

$$(-1)^{i+m+1}.$$

After contracting the edge labelled i, we relabel the remaining edges by preserving their relative order: labels smaller than i remain unchanged, while each label greater than i is decreased by 1.

4 Two Families of Graphs

The proof is organized around two explicit families of graphs. The family $V_N(s)$ describes the quadratic support that remains after eliminating the higher-valence part of the wheel class, while the family $U_N(s)$ provides the auxiliary chains whose differentials perform this elimination. The construction of $V_N(s)$ may be summarized as follows.

Construction of $V_N(s)$ and $U_N(s)$.

1. Input: an odd integer $N = 2m + 1$ and a binary sequence $s = (s_1, \ldots, s_k)$ of choices, with $s_i \in \{\text{left}, \text{right}\}$.

2. Initialize the graph with edges

$$(0,1),\ (0,2),\ (0,3),\ (1,2),\ (2,3).$$

   Denote vertex 1 L, vertex 2 C and vertex 3 R.

3. For $i = 1, \ldots, k$:

   (a) Add a new vertex attached to the current centre vertex C, and denote the new vertex by `new_C`. The new edge (`C`, `new_C`) gets the label $4i + 1$.

   (b) Add an edge (`new_C`, R) (if s_i = left, we instead add the edge (`new_C`, L)). This edge gets the label $4i + 2$.

   (c) Add a second new vertex denoted `new_R` (or `new_L`). Add an edge (`R`, `new_R`) (or (`L`, `new_L`)). This new edge gets the label $4i + 3$.

   (d) Add an edge (`new_C`, `new_R`) (or (`new_C`, `new_L`)). This edge gets the label $4i + 4$.

   (e) Denote the vertices `new_C` and `new_R` (or `new_L`) by C and R (or L) respectively.

   After these k steps, the graph has $2k + 4$ vertices and $4k + 5$ edges.

4. (Only for $U_N(S)$) Add one further centre vertex `new_C` and an edge (`C`, `new_C`). This edge gets the label $4k + 5$. Denote the new vertex C.

5. Now, for both $V_N(S)$ and $U_N(S)$, let $m = N - 2k - 3$ be the number of remaining vertices to be added, and let n denote the current largest vertex label. If $m = 0$, add the final edge (L, R), with edge label $2N - 1$ (This case will only be considered for $V_N(S)$). Otherwise, add the remaining vertices and edges as an arc of triangles as follows: Let us denote the least unassigned edge label $e = 4k + 5$ for $V_N(S)$ and $e = 4k + 6$ for $U_N(S)$.

   (a) Add $(L, n + 1)$ with edge label e.

   (b) For $i = 1, \ldots, m - 1$: Add $(n + i, C)$ with edge label $e + 2i - 1$, and $(n + i, n + i + 1)$ with edge label $e + 2i$.

   (c) Add $(n + m, R)$ with edge label $2N - 1$ for $V_N(S)$ and $2N + 1$ for $U_N(S)$.

The construction is illustrated in the following two examples.
We now record some simple structural properties of the graphs constructed above.

Proposition 4.1. *Let S^{op} be the binary sequence obtained from S by interchanging* left *and* right *in every entry. Then the following statements hold.*

1. *The graphs $V_N(S)$ and $V_N(S^{\text{op}})$ are isomorphic. Likewise, $U_N(S)$ and $U_N(S^{\text{op}})$ are isomorphic. If N is odd, then the isomorphism has an even edge permutation.*

2. *The graph $V_N(\emptyset)$ is the wheel graph W_N.*

3. *For a binary sequence S of length $(N - 3)/2$, every vertex of $V_N(S)$ has valence at most 4.*

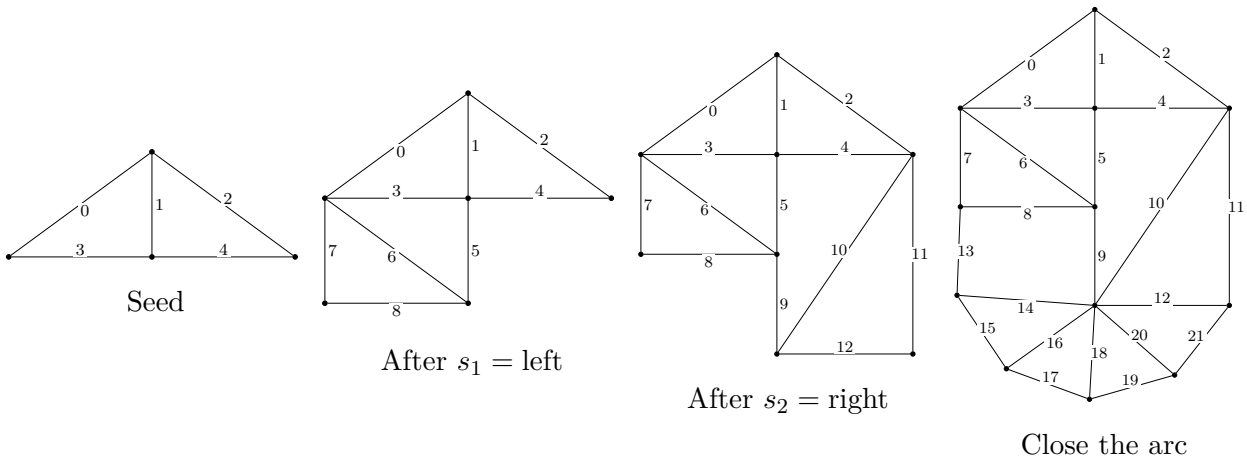


Figure 1: Step-by-step construction of $V_{11}(\text{left}, \text{right})$.

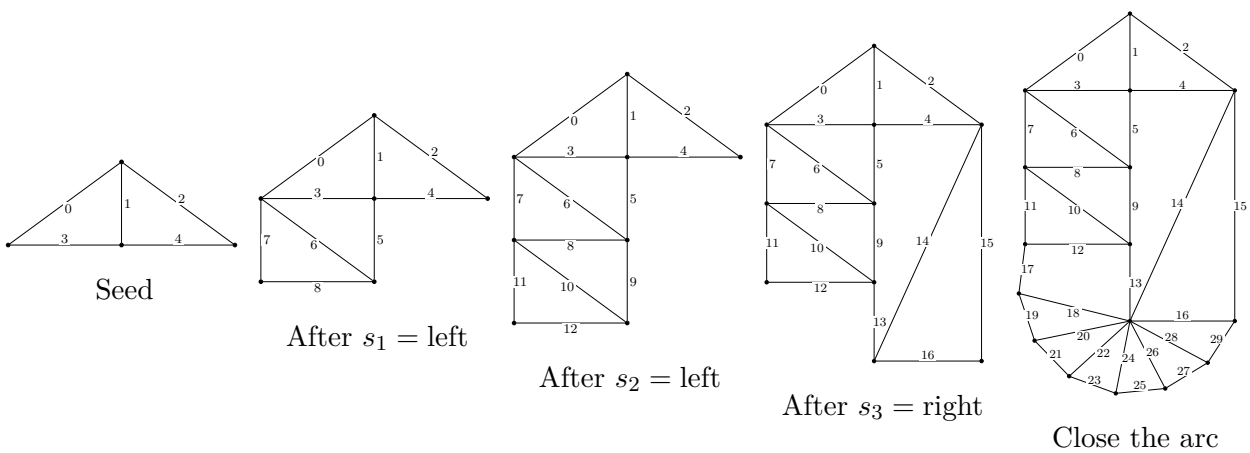


Figure 2: Step-by-step construction of $V_{15}(\text{left}, \text{left}, \text{right})$.

Proof. For item 1, the isomorphism is given by reversing the order in which we add the edges in Steps 2 and 5. Keep in mind that a permutation reversing the order of n items is even if $n \equiv 0, 1 \pmod 4$ and odd otherwise. In Step 2, we add 5 edges, which reverses with two vertex transpositions. In Step 5, we add $2N - 4k - 5$ edges. For $N = 2m + 1$, we have

$$2N - 4k - 5 = 4m - 4k - 3 \equiv 1 \pmod 4,$$

so this reversal is even.

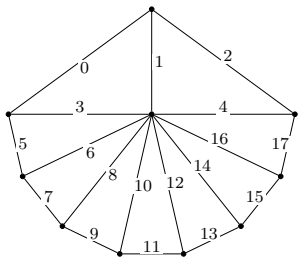
For $V_N(\emptyset)$, we skip from Step 2 directly to Step 5. This results in a wheel graph where vertex 2 is the center hub vertex.

Finally, we note that for any $V_N(S)$, the only vertex which may be more than 4-valent is the vertex denoted C in Step 5. For a sequence of length $(N-3)/2$, we do not have any extra edges to attch to this vertex.

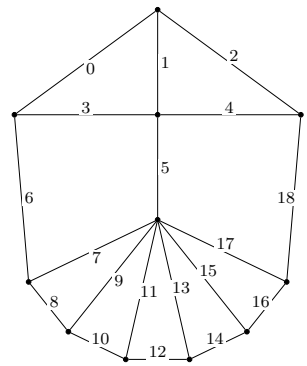
□

Examples of V_N and U_N

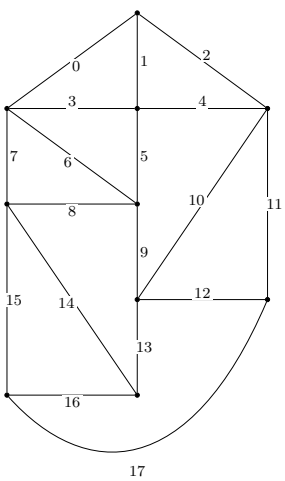
We now collect some examples of graphs in the families $V_N(s)$ and $U_N(s)$.



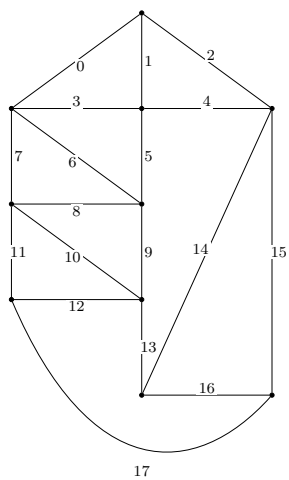
$V_9(\emptyset)$



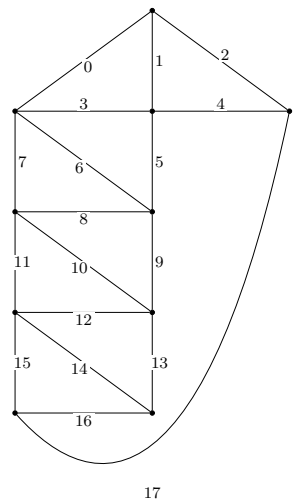
$U_9(\emptyset)$



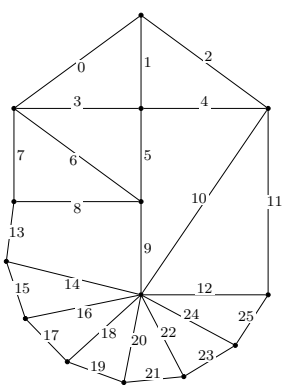
$V_9(\text{left}, \text{right}, \text{left})$



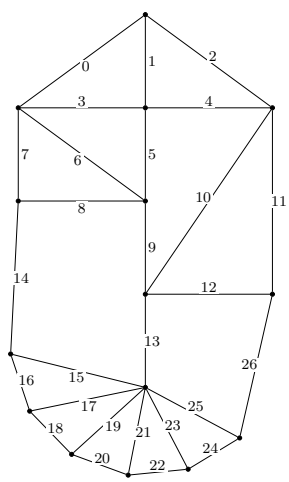
$V_9(\text{left}, \text{left}, \text{right})$



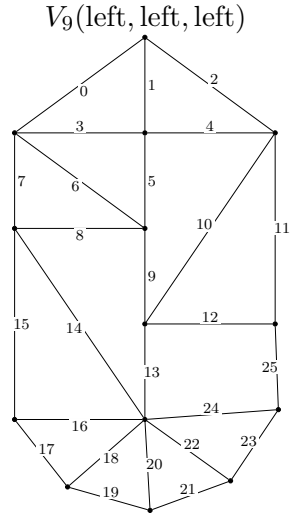
$V_9(\text{left}, \text{left}, \text{left})$



$V_{13}(\text{left}, \text{right})$



$U_{13}(\text{left}, \text{right})$



$V_{13}(\text{left}, \text{right}, \text{left})$

Figure 3: Additional examples in the families $V_N(s)$ and $U_N(s)$.

The families $V_N(s)$ and $U_N(s)$ constructed in this way will be used to produce explicit low-valence representatives for the odd wheel classes.

5 Differential of $U_N(s)$

Let us now consider the edge contraction differential on $U_N(s)$.

The aim of this section is to prove the following Lemma.

Lemma 5.1. *Assume that N is odd. Let $\mathcal{S}_k$ denote the set of all binary sequences of choises $\{\text{left}, \text{right}\}$ of length k. Then*

$$d\left(\sum_{S\in\mathcal{S}_k} U_N(S)\right) = \sum_{T\in\mathcal{S}_{k+1}} V_N(T) - \sum_{S\in\mathcal{S}_k} V_N(S).$$

Proof. For each $U_N(S)$ there are only a handful of edges that may be contracted without creating a double edge, which would kill the resulting graph in the even graph complex.

The edges that may be contracted are the following.

1. The edge labelled $4k+5$, created in Step 4. The resulting graph is identical to $V_N(S)$. The sign of this contraction is
$$(-1)^{4k+5+2N+1+1} = -1.$$

2. The edge labelled $4k+6$, created in Step 5a. This creates a graph that is identical to $V_N(S, \text{left})$. The sign of this contraction is $+1$.

3. The edge labelled $2N+1$, created in Step 5c. This creates a graph that is isomorphic to $V_N(S, \text{right})$. The sign of this contraction is $+1$.

   The isomorphism is given by swapping the edges labelled $2N-1$ and $2N-3$ and by the three insertions
$$2N-3 \mapsto 4(k+1)+1,$$
$$2N-2 \mapsto 4(k+1)+2,$$
$$2N-1 \mapsto 4(k+1)+3.$$
   All these four permutations carry odd sign, so the composed sign is even.

4. Finally, some edges $4i+1$, for $i = 2, \ldots, k$, created in Step (a), may be contracted without creating a double edge. Let us denote such a graph by $U_N^i(S)$.

Consider
$$U_N^i(s_1, \ldots, s_{i-1}, s_i, \ldots, s_k)$$
for a sequence $S = (s_1, \ldots, s_k)$. If $s_i = s_{i-1}$, then $U_N^i(S)$ vanishes because it contains a double edge.

Otherwise, the graphs
$$U_N^i(s_1, \ldots, s_{i-2}, \text{left}, \text{right}, \ldots, s_k)$$
and
$$U_N^i(s_1, \ldots, s_{i-2}, \text{right}, \text{left}, \ldots, s_k)$$
are isomorphic, where the isomorphism is given by the three edge transpositions coming from (b), (c), and (d) of Step 3 in the construction list above. Therefore,
$$U_N^i(s_1, \ldots, s_{i-1}, s_i, \ldots, s_k) = -U_N^i(s_1, \ldots, s_{i-2}, s_i, s_{i-1}, s_{i+1}, \ldots, s_k).$$

Hence these terms cancel pairwise in the sum over all sequences. It follows that

$$\begin{aligned}\sum_{S\in\mathcal{S}_k} d(U_N(S)) &= \sum_{S\in\mathcal{S}_k}\Big(V_N(S,\text{left}) + V_N(S,\text{right}) - V_N(S)\Big)\\ &= \sum_{T\in\mathcal{S}_{k+1}} V_N(T) - \sum_{S\in\mathcal{S}_k} V_N(S).\end{aligned}$$

□

6 Proof of the Main Theorem

It is now time to prove Theorem 2.2. Let us first make a more precise statement, in which we define U.

Theorem 6.1. *For $k \geq 1$, let $\mathcal{S}_k^{\text{left}} \subset \mathcal{S}_k$ denote the subset of binary sequences of length k whose first entry is* left*. For each odd $N \geq 5$, define*

$$U_N^{(0)} := U_N(\emptyset),$$

and

$$U_N^{(k)} := 2\sum_{S\in\mathcal{S}_k^{\text{left}}} U_N(S), \qquad k = 1,\ldots,\frac{N-5}{2},$$

and

$$U_N := \sum_{k=0}^{(N-5)/2} U_N^{(k)} = U_N(\emptyset) + 2\sum_{k=1}^{(N-5)/2}\sum_{S\in\mathcal{S}_k^{\text{left}}} U_N(S).$$

Then

$$W_N + d(U_N) = 2\sum_{S\in\mathcal{S}_{(N-3)/2}^{\text{left}}} V_N(S).$$

By Proposition 4.1, item 3, every graph on the right-hand side has only 3- and 4-valent vertices.

Proof. By Proposition 4.1, item 1,

$$U_N = \sum_{k=0}^{(N-5)/2}\sum_{S\in\mathcal{S}_k} U_N(S).$$

By Lemma 5.1,

$$d(U_N) = \sum_{k=0}^{(N-5)/2}\left(\sum_{T\in\mathcal{S}_{k+1}} V_N(T) - \sum_{S\in\mathcal{S}_k} V_N(S)\right) = \sum_{S\in\mathcal{S}_{(N-3)/2}} V_N(S) - V_N(\emptyset).$$

By Proposition 4.1, item 2, the graph $V_N(\emptyset)$ is the wheel graph W_N. Hence

$$W_N + d(U_N) = \sum_{S\in\mathcal{S}_{(N-3)/2}} V_N(S).$$

Again by Proposition 4.1, item 1,

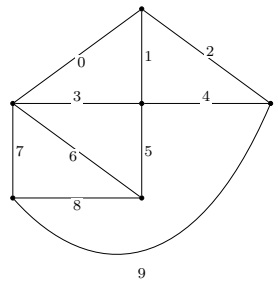
$$\sum_{S\in\mathcal{S}_{(N-3)/2}} V_N(S) = 2\sum_{S\in\mathcal{S}_{(N-3)/2}^{\text{left}}} V_N(S).$$

This proves the theorem above. □

7 Examples

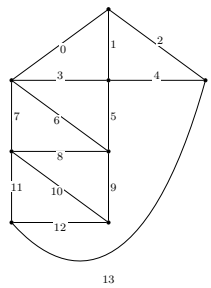
Finally we shall present visualisations of the explicit 3- and 4-valent representatives of the homology classes W_5, W_7, W_9, and W_11, on the form that exists due to Theorem 6.1:

$W_5 \cong$



V_5(left)

$W_7 \cong$



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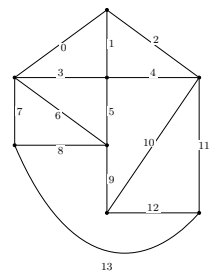
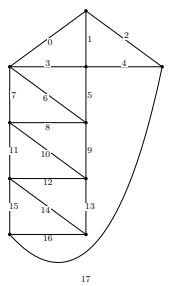


$V_7(LL)$ $V_7(LR)$

$W_9 \cong$



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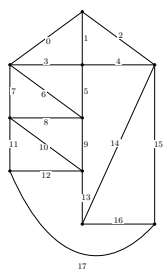

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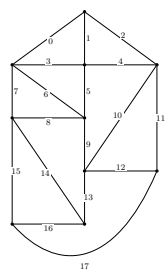

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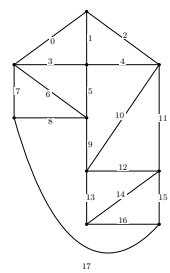
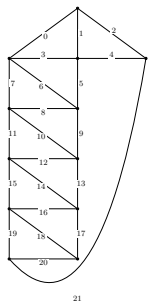
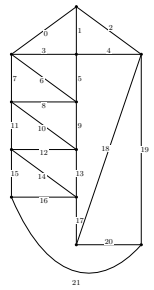
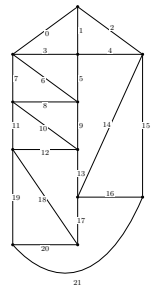
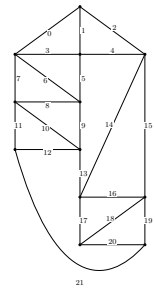
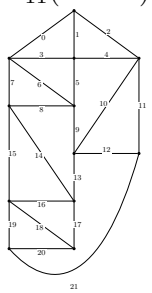
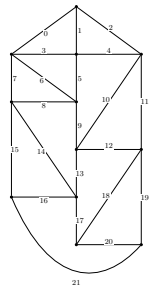
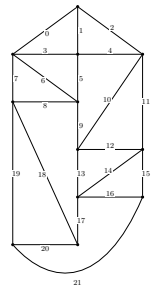
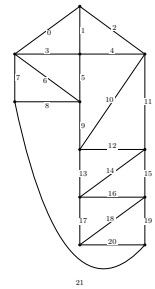


$V_9(LLL)$ $V_9(LLR)$ $V_9(LRL)$ $V_9(LRR)$

$W_{11} \cong$  $+$  $+$  $+$ 

$V_{11}(LLLL)$ $V_{11}(LLLR)$ $V_{11}(LLRL)$ $V_{11}(LLRR)$

$+$  $+$  $+$  $+$ 

$V_{11}(LRLL)$ $V_{11}(LRLR)$ $V_{11}(LRRL)$ $V_{11}(LRRR)$

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